

And what are the specific tools used to compare the parameters Sigma, Phi and Mu in the literature what are the standard ways of comparing two points in time with diferent variables Phi mu sigma

Comparing σ , ϕ , μ in stochastic volatility models: tools used in the literature

In stochastic volatility (SV) and time-varying parameter (TVP) models, σ (volatility scale), ϕ (persistence), and μ (long-run mean) are rarely compared “by eye”. Instead, studies compare their **posterior distributions, credible intervals, and model evidence** across models or time periods.

Bayesian estimation of σ , ϕ , μ

- **MCMC / Gibbs / Metropolis**: Core tools to obtain joint posterior draws of σ , ϕ , μ and latent volatilities, then summarize them with posterior means, standard deviations and credible intervals for each parameter, often in tables for different model variants or datasets (Jacquier et al., 1994; Meyer & Yu, 2000; Kastner, 2016; Chen et al., 2010; Ahsan & Dufour, 2025; Shapovalova, 2021; Anjikwi & Danjuma, 2018).
- **Particle MCMC / RMHMC / variational Bayes / INLA**: Alternative Bayesian methods to approximate the posterior of σ , ϕ , μ and compare how well each method recovers “true” parameter values in simulations (Shapovalova, 2021).

Posterior-based comparison across time or models

Typical comparisons between “two points in time” or “two specifications” use:

- **Posterior means and credible intervals** for σ , ϕ , μ estimated on different subsamples (e.g., pre- and post-regime, or different regimes in Markov-switching models) and checking if intervals overlap (Makatjane & Molefe, 2020; Ghezal et al., 2024; He & Lin, 2022; Anjikwi & Danjuma, 2018).
- **Posterior correlations** to assess whether σ , ϕ , μ change independently or together (Anjikwi & Danjuma, 2018).

Example forms of parameter comparison

Comparison goal	Tool / criterion	Citations
Are σ , ϕ , μ time-varying or constant?	Bayes factors via Savage–Dickey ratio between TVP-SV and constant-parameter models (Chan, 2018).	(Chan, 2018; Chan & Eisenstat, 2018; Chan, 2022)
Which SV structure fits best (different σ , ϕ processes)?	Marginal likelihoods / Bayes factors , Deviance Information Criterion (DIC) across SV specifications (Chan & Eisenstat, 2018; Chan, 2022; G., 2022).	(Chan & Eisenstat, 2018; Chan, 2022; G., 2022)
Is volatility near unit root ($\phi \approx 1$)?	Posterior mean of ϕ and its credible interval , checking if 1 lies in the interval (Ahsan & Dufour, 2025; Anjikwi & Danjuma, 2018).	(Ahsan & Dufour, 2025; Anjikwi & Danjuma, 2018; Kastner, 2016)

FIGURE 1 Common tools to compare SV parameters across models or periods

Structural breaks, regimes, and “two times”

To compare parameters between different periods or regimes:

- **Markov-switching / regime-switching SV models:** σ , ϕ , μ become regime-dependent; inference compares their posterior summaries across regimes (e.g., high vs low volatility) (Ghezal et al., 2024; He & Lin, 2022; Makatjane & Molefe, 2020).
- **Explicit structural break detection** (ICSS, change-point tests, FDA-based tests): first detect breaks in volatility, then estimate SV or ARCH models separately in each segment and compare parameter estimates/intervals across segments (Luo & Chen, 2018; Chen et al., 2010; Kokoszka et al., 2024; Abdullah et al., 2025).

Bayesian hypothesis testing and intervals

- **Bayes factors:** model-level comparison for hypotheses like “ ϕ is constant over time” vs “ ϕ varies over time” (Chan, 2018).
- **Credible / HPD intervals and evidence intervals:** intervals that both estimate likely values of σ , ϕ , μ and quantify evidence for hypotheses such as $|\phi_1 - \phi_2| < \epsilon$ across times or models (Kelter, 2022).

Conclusion

In the SV literature, σ , ϕ , and μ are compared mainly through **Bayesian posterior summaries (means, credible intervals), Bayes factors / model evidence, and regime- or break-specific estimates**, rather than simple point comparisons. To compare two time periods, studies either estimate separate models by period or use regime-switching / time-varying parameter frameworks, and then compare the resulting posterior distributions and interval estimates of σ , ϕ , and μ .

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